FROM TREES TO TECH · ACTIVITY TTT-01 · BIOLOGY + ENERGY

MudWatt Bioelectric League

Can your team make mud generate electricity and defend your design with data?

Win condition: Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation.

THE SCIENCE YOU ARE USING

Electrogenic bacteria. Some bacteria already in ordinary soil are electrogenic: as they digest nutrients, they push spare electrons onto surfaces outside their cells. Bury a graphite anode in oxygen-free mud and these microbes coat it, feeding it a steady stream of electrons.

Completing the circuit. Electrons leave the anode, run through the external wire and a 100 kohm resistor, and reach the cathode resting in the air on top. There they join oxygen and protons from the mud to form water. A multimeter reads the voltage across the resistor, which by Ohm's law tracks the current the microbes produce. More food and more healthy microbes mean more electrons per second, so the measured voltage rises. Power is roughly voltage times current.

HOW TO BUILD AND RUN IT

01 **Inoculate the mud.** Gloves on, working over the tray. Pack the clear 24 oz deli container with sponge-damp pond mud or soil. The bacteria you need are already in the dirt, so no culture is required.

02 **Place the electrodes.** Bury one 4×4 in carbon-felt square near the bottom as the anode and smooth mud over it so no air reaches it. Rest the second 4×4 in carbon-felt square flat on top as the cathode, exposed to the air. Keep the two squares from touching, and run an alligator lead off each.

03 **Add a measured fuel.** Mix in one measured sugar cube (or another approved fuel in a measured amount). Record exactly how much. This is the food that drives the electron output.

04 **Connect and read a baseline.** Wire the 100 kohm resistor in line between the electrodes. Set the multimeter to the DC 200mV range and read the voltage across the resistor (red probe to the cathode side, black to the anode side). Log the first reading. Day one is often only tens of millivolts; the biofilm needs days to grow.

05 **Control one variable.** Pick ONE thing to test all week: fuel type, moisture, or anode contact. Change only that, predict the effect, and compare to baseline.

Safety: Gloves required and build over a tray. No food or drink near the mud bench. Keep the two carbon-felt electrodes from touching. Wash hands after handling soil. No goggles needed; the cell is only a few hundred millivolts. Allergy: Do not use nut-containing food scraps. The standard fuel is a measured sugar cube; banana peel, stale bread, or leaf litter are alternatives.

MATERIALS

Carbon felt, 4×4 in squares	2 per cell (8 for 4 teams, plus spares)
Clear 24 oz deli container	1 per cell (4, plus spares)
100 kohm resistor and alligator leads	1 resistor and 3 to 4 leads per cell
Multimeter (DC 200mV)	1 per team (4, plus 2 spare)
Pond mud or soil	fill each container
Nitrile gloves and a containment tray	per student, 1 tray per cell
Approved fuels	measured sugar cube (banana, bread, or leaf litter)
Labels and tape	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________

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The ONE variable we are testing (fuel type, fuel amount, moisture, or anode packing): ______________________

Baseline reading (mV, day 1): ______________________

Daily readings and our peak (mV): ______________________

DEFEND YOUR DESIGN

What evidence made you change your design? ______________________

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If you ran it again, what is the ONE thing you would change? ______________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Peak voltage reading	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	**100**